ENVIRONMENTAL GEOHAZARDS ENGINEERING · LECTURE

Volcano Monitoring with Remote Sensing

Watching the ground swell, the lava glow and the gas and ash spread, from space

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Computed fringes on a simulated volcano

GOAL OF THIS LECTURE

Use satellites to spot a volcano waking up, follow an eruption as it happens, and measure what it leaves behind.

Ground deformation from InSAR, heat from thermal infrared, gas and ash from ultraviolet and infrared sensors, and case studies from Hawai‘i, Iceland and the Canary Islands.

THIS LECTURE

Outline

01 Volcanoes and why we monitor them
Hazards, warning signs, India’s volcano, and which satellite sees what

02 The ground moves
Magma chambers and dykes, InSAR at every volcano, and combining signals

03 Heat and lava
Thermal infrared, hot-spot systems, and lava volumes from elevation models

04 Gas and ash
Sulfur dioxide from space, finding ash, and keeping aircraft safe

05 Case studies
Kīlauea 2018, Eyjafjallajökull 2010, La Palma 2021 and Grindavík 2023

06 Engineering practice
Where engineers use volcano data, a workflow, limits and exercises

01

PART 1

Volcanoes and why we monitor them

Hazards, warning signs, India’s volcano, and which satellite sees what

VOLCANOES

What volcanoes threaten

Lava flows

Usually slow enough to walk away from, but they destroy everything in their path.

Pyroclastic flows

Avalanches of hot ash and gas, faster than 100 km/h: the deadliest hazard.

Ash fall

Collapses roofs, harms lungs and crops, cuts power and water, and stops jet engines.

Lahars

Volcanic mudflows that race down river valleys, far from the volcano.

Gases

Sulfur dioxide and carbon dioxide: air quality, acid rain, even climate.

Collapse and tsunamis

A flank or summit can fail; Hunga Tonga’s tsunami crossed the Pacific (Lecture 11).

VOLCANOES

Signs that magma is moving

Ground deformation

Magma pushes the ground up and out by centimetres to metres.

Seen from space

Heat

Hot rock, lava and gas warm the surface.

Seen from space

Gas

Sulfur dioxide rises as magma nears the surface.

SO_2 from space; CO_2 hardly

Earthquakes

Rock breaks as magma forces its way up.

Needs seismometers

46%

Of volcanoes that were deforming went on to erupt

6%

Of volcanoes with no deformation erupted

From 540 volcanoes watched by satellite InSAR for about three years (Biggs et al., 2014). Deformation is a strong warning, but not a certain one.

VOLCANOES

India’s volcano: Barren Island

1787
First recorded eruption, Andaman Sea

159 yr
Quiet from 1832, then awake again in 1991

Sep 2025
Two small eruptions, the latest activity

Watched from space

Uninhabited and far from the mainland: satellite heat alerts and radar are the main tools.

Narcondam

A second volcanic island to the north, quiet in historical times.

Deccan Traps

Lava flows 66 million years old: the basalt under much of Maharashtra.

Sources: Global Volcanism Program; Down To Earth; news reports, September 2025.

VOLCANOES

Which satellite sees what

Signal	Satellites	How often
Ground movement	Sentinel-1, ALOS-2, NISAR (radar)	Every 6 to 12 days
Heat	MODIS, VIIRS; Sentinel-2, Landsat	Daily; every few days
Sulfur dioxide	Sentinel-5P (TROPOMI)	Daily
Ash clouds	Himawari, GOES, Meteosat, INSAT-3D	Every 10 to 30 minutes
Lava volume	TanDEM-X, Pléiades stereo, drones	Days to months

NISAR, launched by NASA and ISRO from Sriharikota on 30 July 2025, adds L-band radar, which sees through vegetation on forested volcanoes.

02

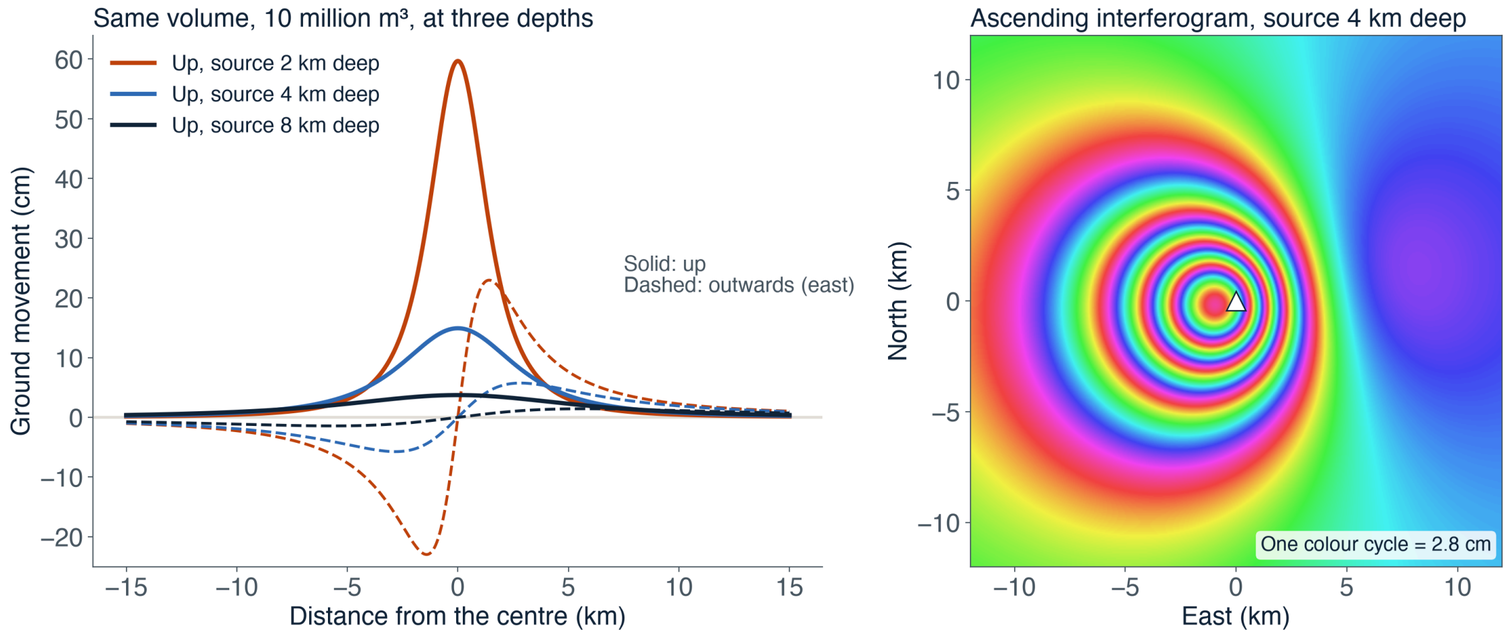
PART 2

The ground moves

Magma chambers and dykes, InSAR at every volcano, and combining signals

DEFORMATION

A swelling magma chamber



Computed: Mogi (1958) point source, Poisson's ratio 0.25.

$$\text{Uplift at centre} = (1 - \nu)\ \Delta V \div (\pi\ d^2)$$

10 million m³ lifts the centre 60 cm at 2 km depth, 15 cm at 4 km and 4 cm at 8 km. The height gives the volume; the width gives the depth.

DEFORMATION

InSAR at every volcano

1,000+
Volcanoes with automatic Sentinel-1 interferograms (COMET LiCSAR)

6–12 days
Between Sentinel-1 images

2.8 cm
Movement per fringe at C-band

Free
Data and processed products

Most volcanoes have no ground instruments: satellites are often first to see unrest.

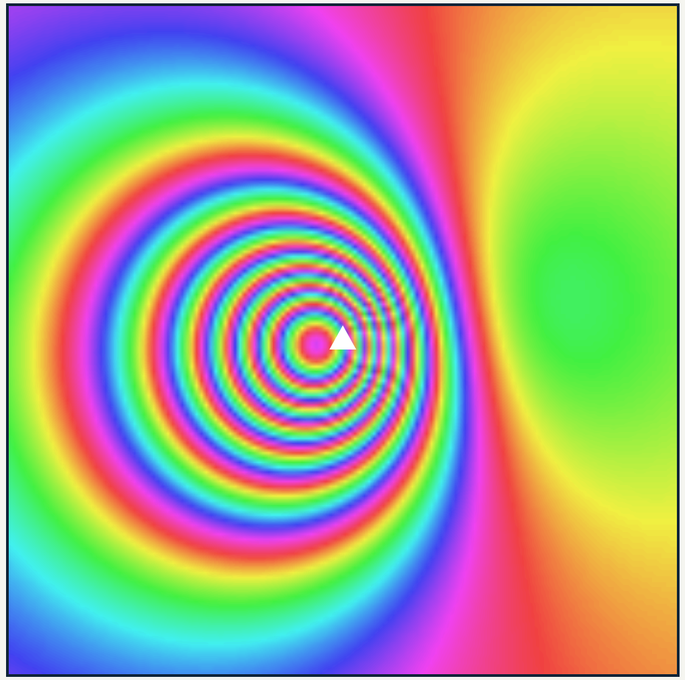
Snow, dense vegetation and water vapour in the air can hide or mimic deformation.

Machine learning now scans thousands of interferograms for new signals.

Sources: Lazecký et al. (2020); COMET Volcano Portal; Biggs et al. (2014).

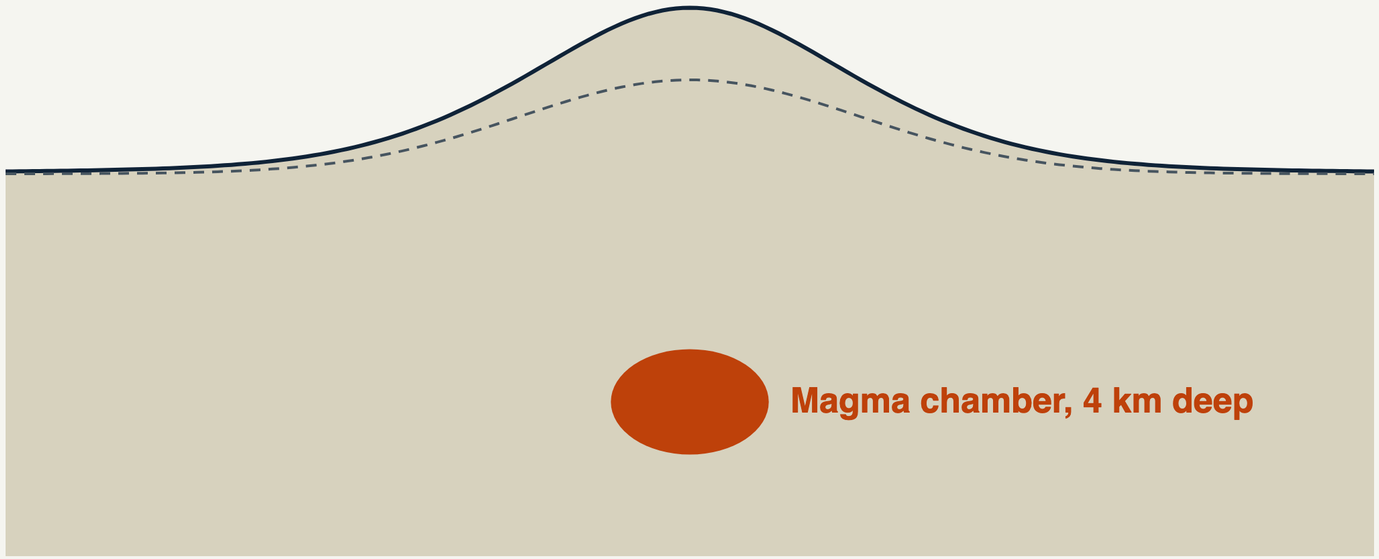
DEFORMATION

Fringes appear as magma arrives



Interferogram, 24 × 24 km

Dashed: ground before. Uplift exaggerated about 5,000 times.



Magma flowing in: ground rises

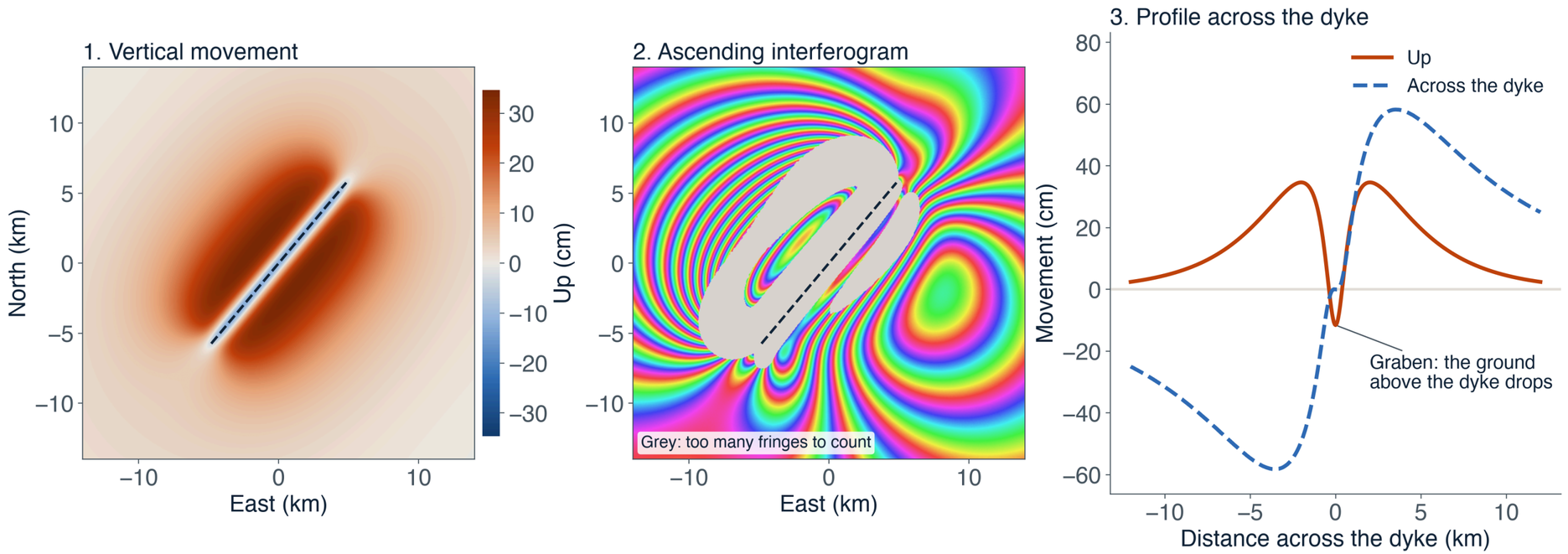
Volume added: 15.7 million m³

Uplift: 23 cm

Fringes: 8

DEFORMATION

Magma on the move: a dyke



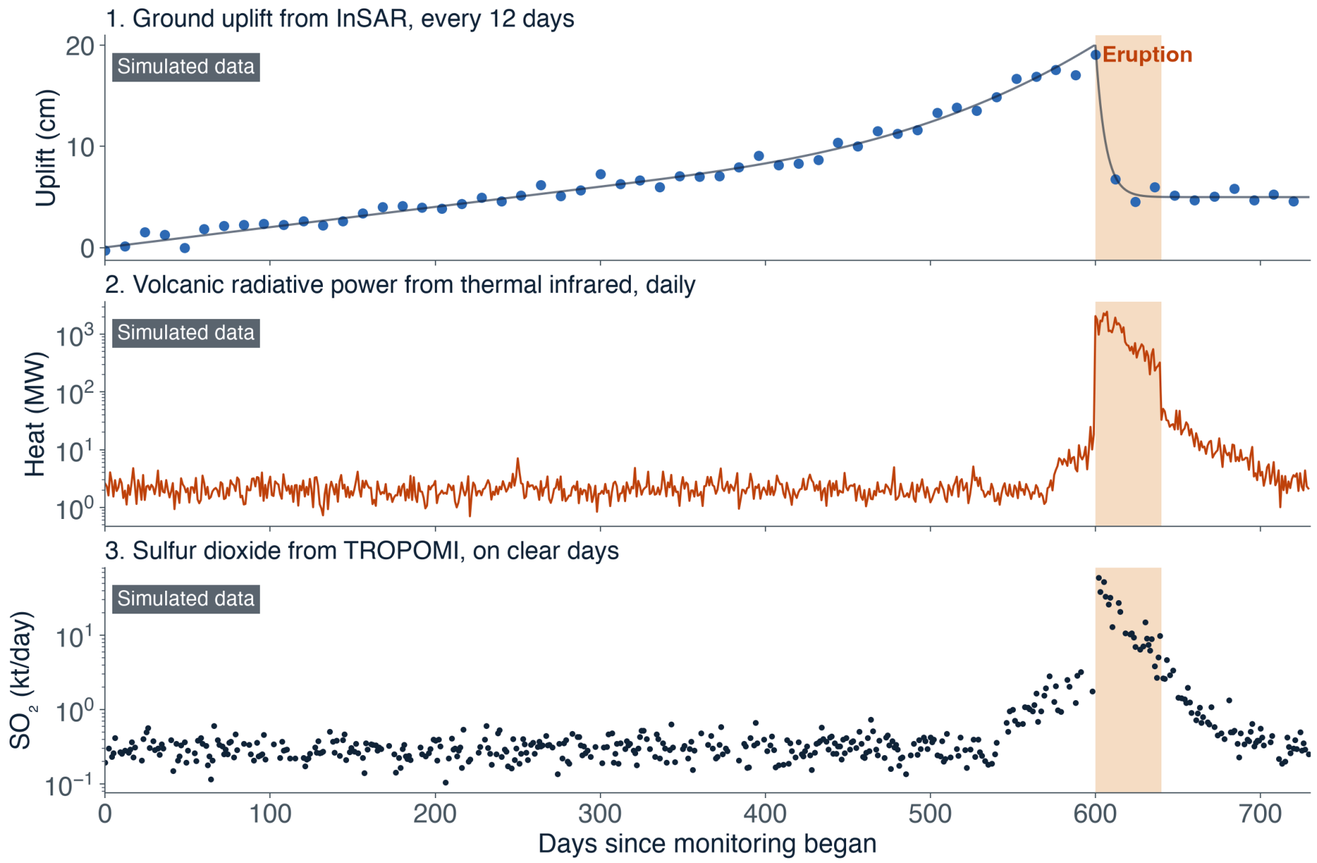
Computed: Okada (1985), a vertical dyke 15 km long, top 0.8 km deep, opening 2 m.

The ground rises on both sides (here 35 cm), drops in a narrow graben above, and is pushed apart by about 1.2 m.

Grindavík, Iceland, 10 November 2023: a dyke about 15 km long formed under the town in hours. InSAR and GNSS mapped it, and the town was evacuated.

DEFORMATION

Combine the signals



Simulated data from one imaginary volcano.

Uplift grows for months: magma is arriving.

Just before the eruption, heat and gas rise too: magma is near the surface.

At the eruption the ground drops, while heat and gas peak.

Each signal alone can mislead; together they tell a story.

03

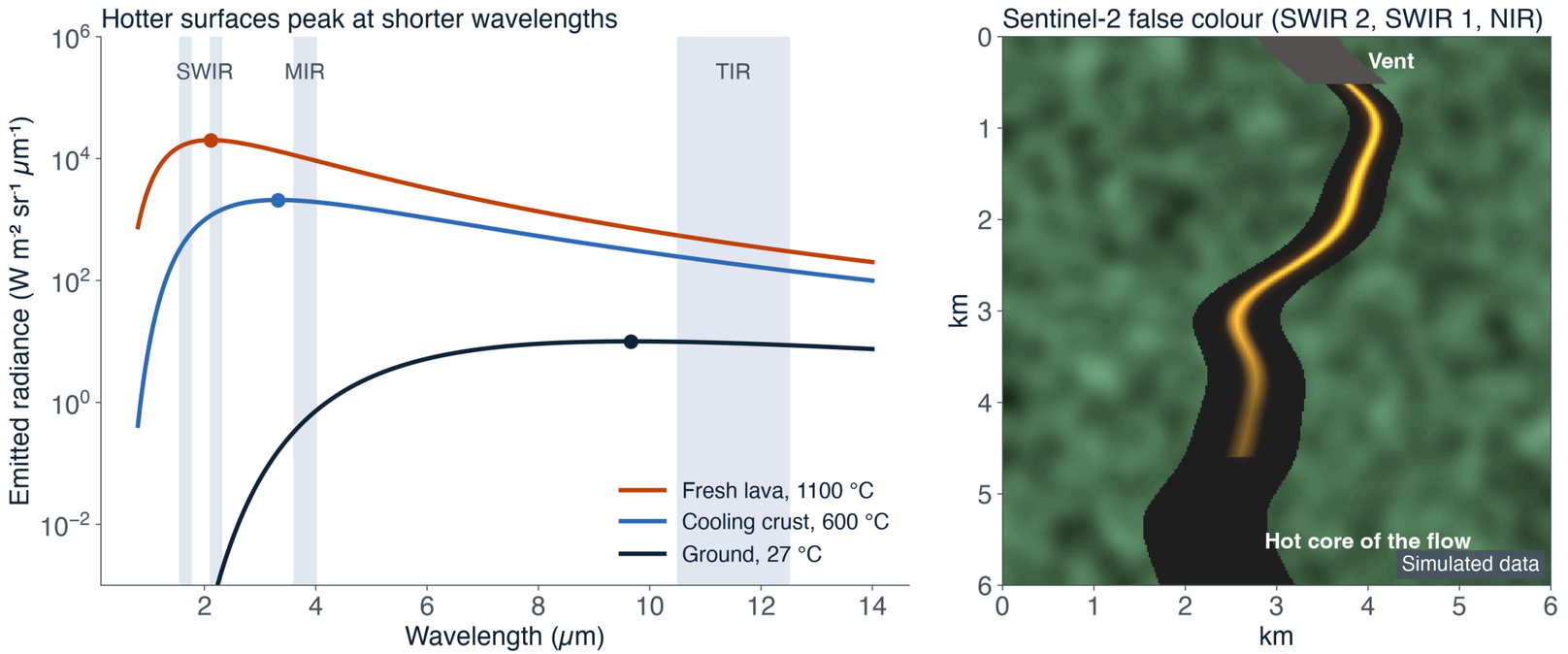
PART 3

Heat and lava

Thermal infrared, hot-spot systems, and lava volumes from elevation models

HEAT

Seeing hot lava



Left: computed from Planck's law. Right: simulated scene.

Fresh lava peaks near 2.1 µm, in the short-wave infrared; cool ground peaks near 9.7 µm.

So Sentinel-2 and Landsat SWIR bands show active lava at 20–30 m, even in daylight; MODIS and VIIRS catch smaller hot spots daily (Lecture 9).

HEAT

Hot-spot systems

MODVOLC

Automatic hot-spot alerts for every volcano from MODIS, since 2000 (Wright et al.).

Sentinel-2 and Landsat

Detailed views of lava flows, lava lakes and hot crater floors, every few days.

MIROVA

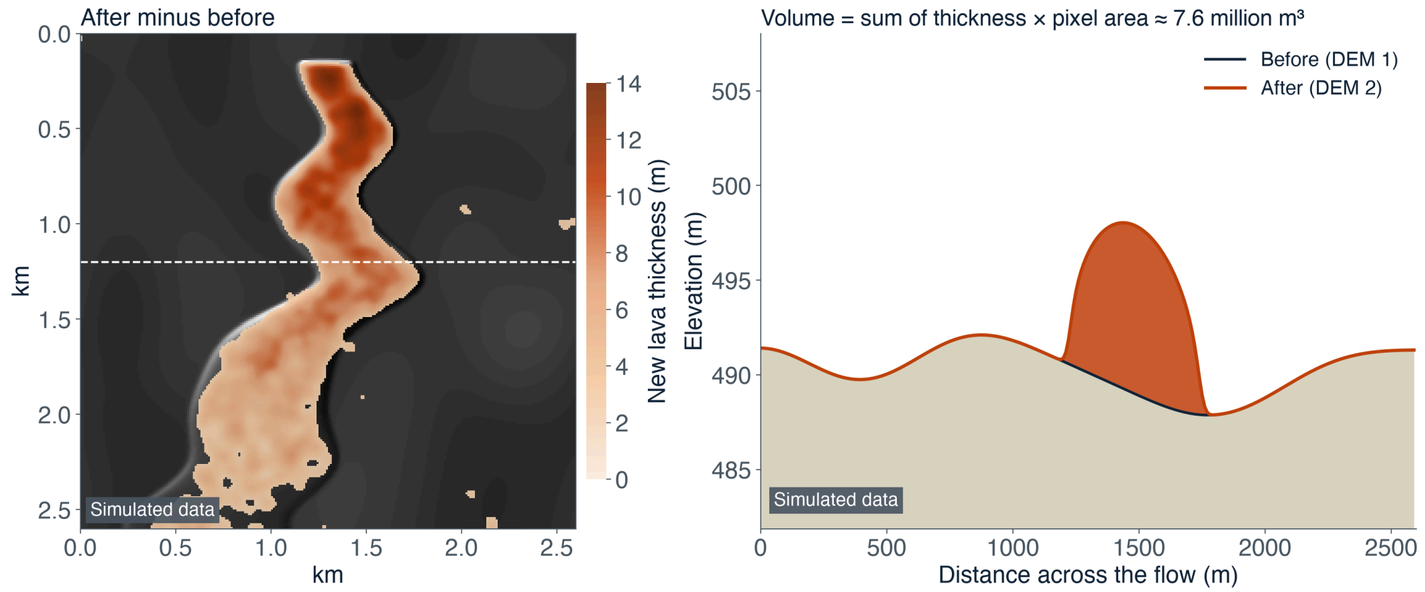
Volcanic radiative power in megawatts: a measure of how much lava is erupting (Coppola et al.).

Heat to lava output

Radiative power can be turned into an approximate eruption rate, for forecasting how far lava will go.

HEAT

How much lava? Subtract elevation models



Simulated: two DEMs with 10 m pixels.

- Thickness = DEM after – DEM before.
- Volume = sum of thickness × pixel area: here about 7.6 million m³.
- DEMs come from TanDEM-X radar, Pléiades stereo images or drones.
- Each DEM has errors of about a metre: ignore changes thinner than that.

04

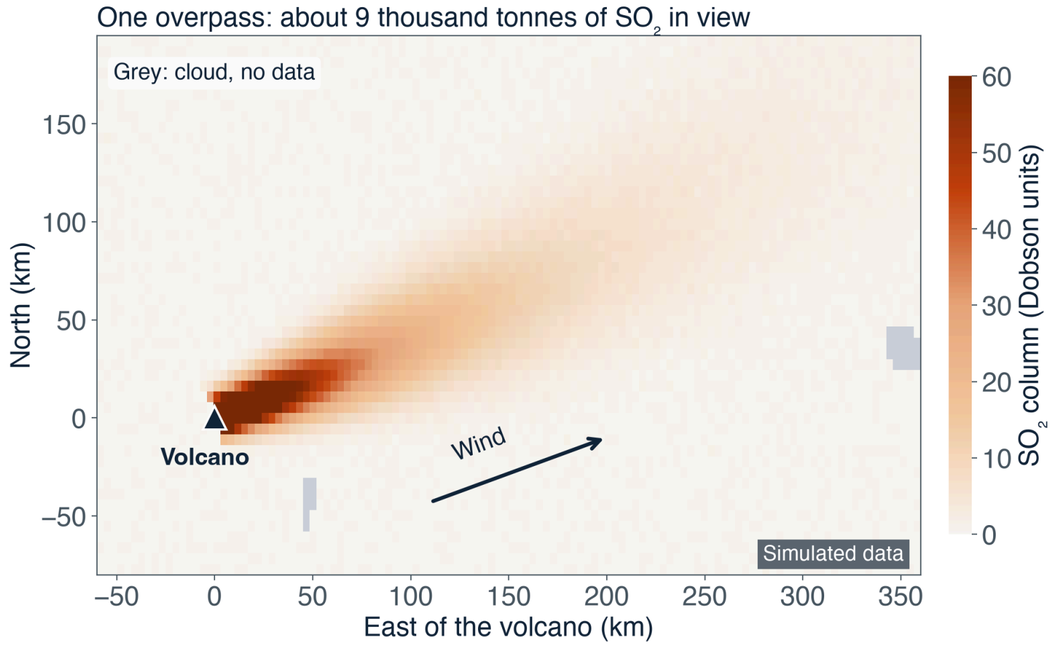
PART 4

Gas and ash

Sulfur dioxide from space, finding ash, and keeping aircraft safe

GAS AND ASH

Sulfur dioxide from space



Simulated: one TROPOMI-like overpass.

SO_2 absorbs ultraviolet sunlight; TROPOMI on Sentinel-5P maps it every day.

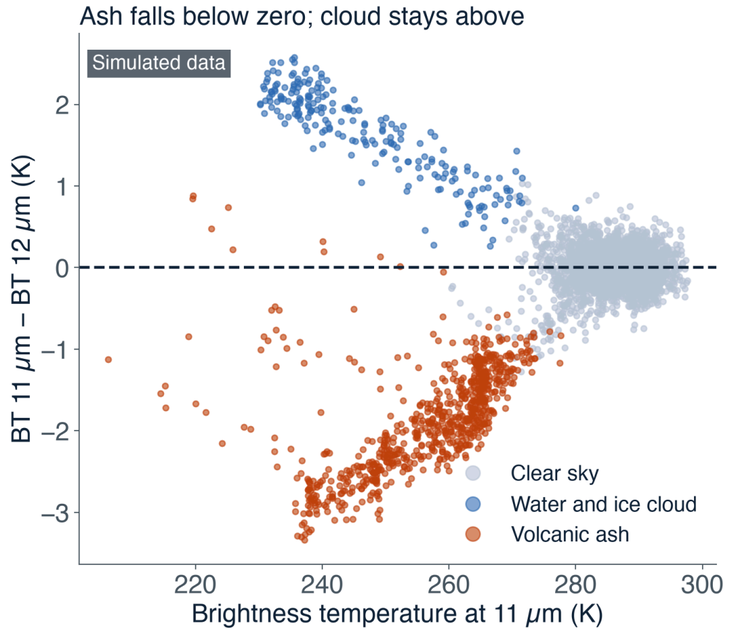
1 Dobson unit = 2.86×10^{-5} kg of SO_2 per m²: add up the pixels for the mass.

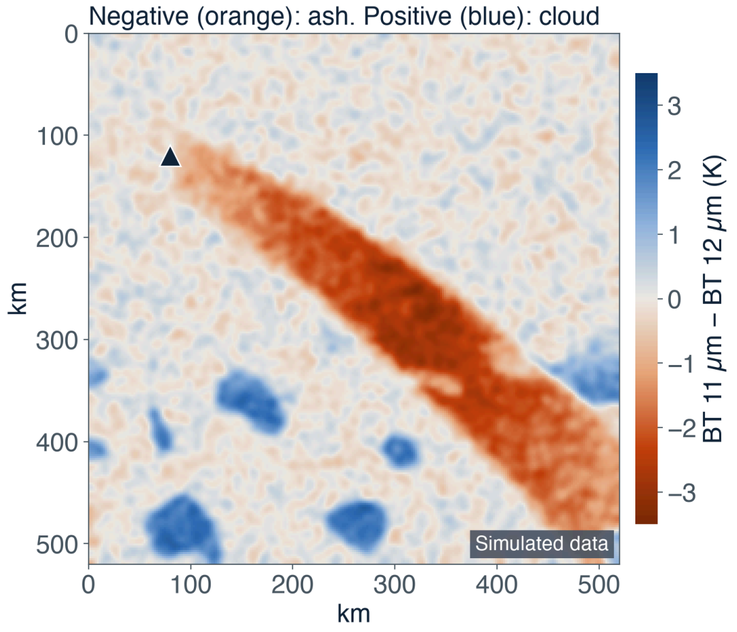
Rising SO_2 can mean fresh magma near the surface.

Pinatubo, 1991: about 15–20 million tonnes of SO_2 cooled the world by about 0.5 °C.

GAS AND ASH

Finding ash: the split-window test





Simulated scene.

Ash absorbs more at 11 µm than at 12 µm; water and ice do the opposite. So **BT11 – BT12 below zero** flags ash (Prata, 1989).

Geostationary weather satellites repeat every 10 to 30 minutes, so they can follow an ash cloud as it drifts.

GAS AND ASH

Keeping aircraft out of ash

9
Volcanic Ash Advisory Centres around the world

57 km
Hunga Tonga plume, 2022: a record

10–30 min
Between geostationary images

≈ 1,100 °C
Roughly where ash melts; jet engines run hotter

Melted ash coats engine parts as glass and can stop the engines.

Advisory centres combine satellite images with wind models to forecast where ash goes.

Plume height comes from cloud-top temperature, shadows, or the parallax between two satellites.

Sources: ICAO International Airways Volcano Watch; Proud et al. (2022), Science.

05

PART 5

Case studies

Kīlauea 2018, Eyjafjallajökull 2010, La Palma 2021 and Grindavík 2023

CASE STUDY

Kīlauea, Hawai‘i, May–August 2018

716
Homes destroyed by lava

≈ 1 km³
Of lava erupted from 24 fissures

≈ 0.8 km³
Volume lost as the summit caldera collapsed

60,000
Earthquakes during the eruption

InSAR and GNSS showed the summit sinking as magma drained to the rift zone.

Thermal images and drones tracked the lava; sulfur dioxide spread vog across the islands.

Practical 1 processes Sentinel-1 data for this eruption in SNAP.

Sources: USGS Hawaiian Volcano Observatory; Hawai‘i County.

CASE STUDY

Eyjafjallajökull, Iceland, April 2010

107,000
Flights cancelled

10 million
Passengers stranded

23
Countries with closed airspace at the peak

$1.7 bn
Lost by airlines

A modest eruption under ice: meltwater shattered the magma into fine, glassy ash.

Satellites and models tracked the ash cloud across Europe for weeks.

Afterwards Europe moved from closing all airspace to limits on ash concentration.

Sources: IATA; Eurocontrol; Global Volcanism Program.

CASE STUDY

La Palma, Canary Islands, 2021

85 days
19 September to 13 December 2021

2,988
Buildings destroyed (Copernicus EMS)

1,237 ha
Covered by lava

7,000+
People evacuated

Ground stations and InSAR saw up to about 20 cm of uplift in the eight days before.

Copernicus EMS mapped the lava front almost daily from satellite and aerial images.

Lava reached the sea and built two new deltas of land.

Sources: Copernicus EMS; ESA; Scientific Reports (2024).

CASE STUDY

Grindavík, Iceland, 2023–2024

3,790
Residents evacuated on 10 November 2023

15 km
Dyke formed under the town in hours

22,000+
Earthquakes from 25 October

Jan 2024
Lava entered the town; three houses burned

InSAR and GNSS mapped the dyke and a graben through the town within a day.

Earth barriers were built to steer lava away from the town and the Svartsengi power plant.

Monitoring guided both the evacuation and the engineering response.

Sources: Icelandic Meteorological Office; Natural Hazards (2026); Reykjavík Grapevine.

06

PART 6

Engineering practice

Where engineers use volcano data, a workflow, limits and exercises

ENGINEERING

Where engineers use volcano data

Hazard zoning

Lava, lahar and ash zones guide land use and building rules.

Lahar paths

Bridges, dams and roads in valleys below volcanoes; Pinatubo’s lahars lasted years.

Ash on roofs

10 cm of wet ash weighs about 160 kg/m²: enough to damage weak roofs.

Lava barriers

Earth walls steered lava around a town and a power plant in Iceland.

Lifelines

Ash shorts power lines, clogs water treatment and closes airports.

Geothermal energy

Volcanic heat makes power; monitoring protects the plants.

ENGINEERING

A volcano monitoring workflow

01
Know the volcano: history and hazards

02
Baseline: InSAR, heat, gas, seismic

03
Detect change: deformation, heat, gas

04
Model the source: depth and volume

05
Set alert levels with the observatory

06
Track the eruption: lava, ash, gas

07
Map impacts: lava, ash, damage

08
Update hazard maps and plans

Free data and tools: COMET LiCSAR, MIROVA, Sentinel-1, -2 and -5P, Copernicus EMS, the Global Volcanism Program.

Student project: a Sentinel-1 interferogram and a Sentinel-2 SWIR image of a recent eruption.

ENGINEERING

Limits and good practice

Limits

- Clouds and plumes hide the ground from optical sensors
- Snow, vegetation and the air disturb InSAR
- Satellite revisits can miss fast changes
- Not all unrest ends in an eruption

Good practice

- Combine deformation, heat, gas and seismic data
- Use radar in all weather; correct for the atmosphere
- Use daily and geostationary sensors for fast events
- Work with the observatory and its alert levels

EXERCISES

Try these

01 A magma chamber 4 km deep gains 10 million m³. Find the uplift at the centre ($\nu = 0.25$). About how many Sentinel-1 fringes is that, ignoring the viewing angle?

02 The same volume enters a chamber 8 km deep. How does the uplift change, and how does the shape of the bulge change?

03 10 cm of wet ash with a density of 1,600 kg/m³ falls on a roof. Find the load in kg/m² and kPa.

04 An SO_2 plume covers 100 km × 20 km with an average of 15 Dobson units. Find its mass (1 DU = 2.86×10^{-5} kg/m²).

EXERCISES

Answers

01 $0.75 \times 10^{7} \div (\pi \times 4{,}000^{2}) \approx$ **0.15 m = 15 cm**; $15 \div 2.8 \approx$ **5 fringes**.

02 Uplift falls with depth squared: **4 times smaller, about 3.7 cm**. The bulge becomes about twice as wide.

03 $0.10 \times 1{,}600 =$ **160 kg/m²**; $\times 9.81 \approx$ **1.6 kPa**: in the damage range for weak, timber-framed roofs.

04 Area 2×10^{9} m²; $15 \times 2.86 \times 10^{-5} \times 2 \times 10^{9} \approx$ **860,000 kg ≈ 860 tonnes**.

RECAP

Key points

01 Magma announces itself: the ground swells, heats up and gives off gas.

02 InSAR now watches every volcano; models turn fringes into depth and volume.

03 Thermal infrared finds lava; elevation models measure how much erupted.

04 UV and infrared sensors track SO_2 and ash, keeping aircraft safe.

05 Combine signals, and work with the observatory.

FURTHER READING

References

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Thank you. Questions?

Figures and the animation were made for this lecture. The magma-chamber (Mogi, 1958) and dyke (Okada, 1985) deformation, the Planck curves and the animation are computed; the multi-sensor records, lava scene, elevation models, SO_2 plume and ash scene are simulated and labelled as such; the cover fringes are computed on a simulated volcano. Case-study facts: USGS, Copernicus EMS, ESA, IATA, Eurocontrol, the Icelandic Meteorological Office, the Global Volcanism Program and the papers on the previous slide.